Quantum Error Correction (QEC) & Readout Error Mitigation (REM) in Superconducting Quantum Computers

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Outline

QEC General-Info

Computers and Errors

Quantum vs Classical Computers:

- Classical & Quantum computers with `n` bits/qubits:
  - Number of possible states $= 2^n$ becuase of the binary digits
  - Units of memory $= n$
- Classical computers use one state (out of 2^n states) at a time
- Quantum computers evolve a superposition over all 2^n states, and can use interference across that (Hilbert) space

$\Rightarrow$ Quantum computer power is about enabling computational speedups!!

Error Correction:

- **Classical error:**
  - Classical bits only have one kind of error: $0 \leftrightarrow 1$.
  - Noise flips them rarely
  - **Error Correction:** copy bits freely

  $\Rightarrow$ Classical bits are more stable!!!
- **Quantum error:**
  - Qubits can be in superposition of $|0\rangle$ or $|1\rangle$: $\alpha |0\rangle + \beta |1\rangle$
  - Interaction with the environment changes the amplitude α or β
  - Classical redundancy is not efficient, **No Clonning Theorem!!**

  $\Rightarrow$ Qubits are less stable

Quantum Errors

Qubits' Errors:

- Bit-Flip (X error)
- Phase-Flip (Z-error)
- Bit-Flip + Phase-Flip (Y-error)
- Amplitude damping
- Decoherence
- Leakage out of computational state space, e.g. $|1\rangle \rightarrow |2\rangle$
- Crosstalk: Microwave pulses intended for one qubit spill over to neighbors
- Readout errors: Qubit measurement uses microwave resonators that could be noisy and imperfect

Noises that produce quantum errors

- Qubits state changes with:
  - temperature
  - interaction with environmental noise
  - imperfect control pulses
- Quantum memory decoherence in microseconds to milliseconds $\Longrightarrow$ errors accumulate fast
- Gates and measurements produce noise
- Long computation, fault-tolerant gates, scalable quantum processors
  $\Longrightarrow$ **Quantum Error Correction (QEC)** is required for any large-scale quantum computation, as quantum computer loses coherence long before it can finish any meaningful algorithm

QuantumErrorCorrection – QEC

Quantum Limitation:

- Quantum information can not be measured directly without disturbing it
- Quantum mechanics forbids copying unknown states (**No-Cloning Theorem**)
- QEC uses **entanglement** and **syndrome measurement** that adds complexity and many more qubits

Quantum error correction is the set of techniques that keep quantum information alive, as the qubits are extremely fragile!

QEC method introduces **logical qubits**

- Encoding many physical qubits using entanglement
- Errors on individual qubits can be detected without measuring their quantum information
- Error types can be identified using Syndrome measurement
- Error can be corrected without disturbing qubits' quantum state.

Ultimate goal: Algorithm that measures error syndromes & information to capture what error occurred without collapsing qubits' quantum state.

QEC Meth

QEC Methodology

QEC Goals:

- **Encoding:** Distribute logical qubit information across a lattice of physical qubits to detect and correct *random* errors without collapsing the underlying quantum state through a continuous, *real-time feedback loop*
- **Non-Destructive Detection:** Utilize local parity checks (syndromes) to identify errors while preserving the *data qubits' superposition*
- **Surpassing the Threshold:** Maintain physical gate fidelity above the *fault-tolerant threshold*, ensuring that increasing the system size exponentially reduces the logical error rate (explained in slide 33)

QEC Methodology:

- **Encoding:** Single logical state *smeared* across a lattice of physical data qubits using entanglement (slide 80 in backup slides)
- **Syndrome Extraction:** Ancilla qubits interact with data qubits to perform Stabilizer Measurements, gathering information about errors
- **Decoding:** Classical hardware (like FPGAs) runs an algorithm to interpret the *syndrome* and identify the most likely physical errors.
- **Correction:** The system applies a "recovery" operation in software rather than a physical gate.

Fidelity

Fidelity ($0 \leq value \leq 1$) in Quantum Computing describes how similar a quantum operation or state is to the one intended:
Fidelity $= 1 \Rightarrow$ Quantum system is perfect
Fidelity $= 0 \Rightarrow$ Quantum system totally random (coin flip)

Fidelity types:

- State Fidelity: Measures how well a prepared state matches the target state
- Gate Fidelity: Computed with Randomized Benchmarking (RB); separates gate errors from State Preparation and Measurement (SPAM) errors
- Readout Fidelity (often the bottleneck): Probability that the recorded classical bit matches the qubits' quantum state: $F_{readout} = 1 - \frac{P(0|1)+P(1|0)}{2}$
- Logical Fidelity (QEC Fidelity): Take physical qubits with Fidelity $> 99\%$ and combine them to make a logical qubit with improved Fidelity, e.g. $\approx 99.9999999\%$

Context	What it measures	Source of Error
State Fidelity	Overlap of two states	T_1 decay + T_2 dephasing
Gate Fidelity	Accuracy of rotation	Over rotation + microwave noise
Readout Fidelity	Reliability of measurement	T_1 decay while measuring
Logical Fidelity	Success of QEC code	Physical errors exhausting the code

QEC and Qubits' Type

Physical qubits, the key elements of Quantum Computers (QC), process data and perform computation, and are categorized into three functional classes:

- **Data-Qubits:** QC's building blocks that store the logical information. Maintained and corrected via QEC codes without collapsing their quantum states
- **Ancilla-Qubits:** The QC's detecting components, entangled with data-qubits to extract syndromes via measurement to correct the data-qubits without collapsing them. Often located between data-qubits, their exact location depends on the topology.
- **Flag-Qubits:** Designed to detect circuit-level noise or fault propagation in the circuit, by spotting the spread of a single ancilla fault that results in high-weight-error $\implies$ to ensure the fault-tolerant operation

Qubits Population: In a common architecture (e.g., Surface code), the data-to-ancilla qubits ratio is roughly 1:1. Total overhead scales with the Error Correcting Code (ECC) and the required distance parameter

Note: Qubits are typically arranged in a 2D square or hexagonal layout, like the **Surface Code** *(slide 24) in the industry standard (Google, IBM)*

QEC Pathway

- **Access QC** data from IBM backend (more in slide 15) and applied cuts on: $T_1 > 50\mu s$ & readout error $< 5\%$
- **Build and identify logical qubits** (slide 25):
  - Calculate *Average 2-Qubit Gate Error* for several logical patches based on selected distance
  - The patch with the lowest average is a good candidate to perform QEC
- **Build noise model from backend** dataset (slide 17):
  - From coherence time (T) & readout time (t_g) compute error probabilities:

$$P = 1 - e^{-\frac{t_g}{T}}$$

  - Combine single-qubit and double-qubits errors into *Depolarizing Channel* model (slide 92)
  - Tool: Use qiskit_aer.noise.NoiseModel to ingest the dataset and create a simulator to represent the logical patch
- **Memory Experiment** Measures the Logical Error Rate (P_L) over time.
  - **Initialization:** Prepare 9 data qubits in a logical $|0\rangle_L$ state.
  - **Syndrome Extraction:** Perform T rounds of X and Z stabilizers (where $T = d$ is the distance parameter)
  - **Final Readout:** Measure all data qubits; use a Decoder (like MWPM) to reconcile the syndromes and determine the final logical state

QEC Pathway – Continue

- **Decoding: The "Brain" of QEC:** *The Challenge* is that the Raw syndrome data only tells an error occurred, not *where*. The most likely source of the error must be inferred.
  - **The Decoder (PyMatching):** Utilizes Minimum Weight Perfect Matching (MWPM) to map syndromes to a graph and find the most probable error chains.
  - **Noise-Aware Decoding (Weighting):** Instead of assuming all qubits are equal, feed the decoder with the hardware-specific error rates (p_i):
    - High-error qubits are assigned **lower weights** ($W \propto \ln[(1-p)/p]$).
    - This "biases" the decoder to check those qubits first, significantly boosting **Logical Fidelity**.

The Result

A "Smarter" decoder can achieve a higher **Error Threshold**, allowing the code to succeed even on noisier hardware.

- **Threshold Analysis** Details in slide 35

QEC Simulation Framework

1. **Architecture & Mapping**
   - **Topology:** Transposed a 3×3 Surface Code ($d = 3$) onto the **IBM Heron** (Heavy-Hex) layout.
   - **Data Extraction:** Parsed T_1, T_2, and gate fidelities from `ibm_fez` via `Qiskit Runtime`.
2. **Noise Modeling (Stim)**
   - **Depolarizing Channel:** Converted physical error rates into *Circuit-Level Noise*.
   - **Measurement Noise:** Modeled noisy syndrome extraction to simulate realistic d rounds of detection.
3. **Decoding & Verification (PyMatching)**
   - **MWPM:** Used Minimum Weight Perfect Matching with weights derived from the hardware calibration DataFrame.
   - **Monte Carlo:** Executed 10^5 shots to calculate the **Logical Error Rate** **(P_L)**.

The link to the GitHub repository that contains the simulation codes: QEC Repo

IBM Backend

Access Dataset – QC Backend

- Cloud-based, full-stack technology providing access to superconducting, transmon-based quantum processors, with systems exceeding 100 qubits
- Uses **Qiskit**, an open-source software stack, allowing users to run experiments, algorithms, and simulations for complex problems
- **Access IBM data:**
  - Create IBM account from IBM webpage and attain an API key
  - Download the CSV file from the free backends or use the online backend data
- Calibrated data are physical parameters and error rates that characterize its current performance
- **Calibration:** the process of measuring qubits' drifts and updating the control pulses to ensure gates remain accurate
- Qubits are prepared and measured 1024+ times (2^{10} classical value, not a physics limit), and the values are reported as the probability of preparing and measuring.

IBM Calibration

Calibration: Automated set of experiments that use Pulse to fine-tune the microwave signals sent to the QPU as follow:

- **Spectroscopy (resonance search):** Find each qubit's resonant frequency by sweeping with microwave frequencies
- **Rabi Experiment:** Changing microwave pulse amplitude to find the precise power needed to rotate the qubit's state by 180 degrees
- **Ramsey/Echo Experiment:** measures decoherence time (T_1, T_2) to correct fine-frequency shifts
- **Error Benchmarking:** to calculate the final average error rates reported in the backend properties using techniques like **Randomized Benchmarking (RB)** (slide 36)

Calibration schedule to balance the uptime accurately. During the calibration, the system is not accessible:

- Hourly calibration: 2-3 minutes to check gates and readout stability
- Daily calibration: 30–90 minute maintenance, full re-optimization of gate amplitudes, phases, and error benchmarking
- As needed: If monitoring detects the qubit's error rate is beyond a certain threshold, system undergoes *local* recalibration of the qubit(s).

IBM Calibration Data

- **Coherence Times:**
  - T_1 or Relaxation Time (μs): How long a qubit stays in the excited state ($|1\rangle$) state before decaying to ground state ($|0\rangle$)
  - T_2 or Dephasing Time (μs): How long a qubit maintains its phase (ϕ) in the superposition: $|\psi\rangle = |0\rangle + e^{i\phi}|1\rangle$
  - To compute probability from coherence time: $P = 1 - e^{-tg/T_1}$ where t_g is the readout time (ns).
  - Microwave pulses affect the coherence time as well (slide 88)
- **Gate Properties:**
  - Single-qubit error rates: The average error for basic gates, e.g., S_X and X (mathematical formalism in slide 90)
  - Two-qubit error rates: Errors for entangling gates (slide 81), typically the highest gate errors in a system; in IBM superconducting devices, the native entangling gate is CZ (not CNOT), because of the SQUID usage (slide 82, 83)
  - Gate durations (ns): The physical time it takes to execute a specific pulse
- **Measurement Metrics:**
  - Readout Error: The probability that a $|0\rangle$ is measured as a $|1\rangle$ ($P(0|1)$), or vice versa ($P(1|0)$):
    - Measurement process takes longer than T_1: $P(0|1)$ is dominant
    - thermal excitation: $P(1|0)$ is dominant
  - Readout assignment error: The average of $P(1|0)$ and $P(0|1)$ to account for both type of measurement errors

Superconducting Transmon

The Superconducting Transmon

- **System Architecture**
  - **Dilution Refrigerator:** To refrigerate and keep qubits to $\approx 20\,\mathrm{mK}$, made with gold plating that ensures high thermal conductivity to qubits' heat sink while operating
  - **Control System:** Classical computers and microwave (MW) pulses to manipulate quantum states
- **Transmon Qubits (Artificial Atoms):**
  - **Artificial Atoms:** Made up of a collective electronic state (Cooper pairs) oscillating in a circuit
  - **Composition:** Superconducting circuit containing a capacitor and a Josephson Junction (non-linear inductor)
  - **Anharmonicity:** The junction creates unequal energy spacing, isolating the $|0\rangle \rightarrow |1\rangle$ transition to create a usable Two-Level System (TLS).
- **Addressing & Frequency:**
  - **Frequency Tuning:** Each transmon has a unique excitation frequency to allow for individual spatial addressing.
  - **Crosstalk Mitigation:** Distinct frequencies prevent accidental operations on neighboring qubits.

Microwave Pulses

The benefits of having an energy gap between ground state and excited state in the microwave regime (3-10 GHz) are:

- Superconducting (artificial) atoms are kept in 20 mK temperature refrigerator. Thus, thermal excitation does not affect the excitation to higher quantum energy levels.

$$\begin{aligned} E_{thermal} &= k_B T = 1.38 \times 10^{-23} \text{ J/K} \times 0.020\text{K} = 2.76 \times 10^{-25}\text{J} \\ \omega_{thermal} &= E_{thermal}\text{J}/h = \frac{2.76 \times 10^{-25}}{6.626 \times 10^{-34} \text{ J} \cdot \text{s}} = 0.4167\,GHz \end{aligned} \tag{1}$$

- In the microwave regime, having a narrow bandwidth is possible. **IMPORTANCE**:
  - TLS single frequency per qubit $\implies$ Prevents **crosstalk** in neighboring qubits;
  - Resonators and amplifiers work well in the microwave regime;
  - Multiple transmon qubits connected via a single-line wire (to avoid overheating the refrigerators) can cause leakage to higher states $\implies$ narrow bandwidth can help avoid it
  - Fast control gates are feasible (more in slide 22)

Design QC in Microwave Regime

QCs are designed so that the TLS change in frequency is in the microwave regime:

$$h\omega = \sqrt{8E_C E_{JJ}} - E_C$$

$$E_C = \frac{e^2}{2C}$$

$$E_{JJ} = \frac{hI}{2e}$$

- Selecting the capacitor's shunt to match its energy to the requested frequency by changing the capacitor pads
- Selecting Josephson Junction inductor to match its energy to the requested frequency by tuning the Junction area and oxide barrier

This will result in many qubits with slightly variant frequencies to fit in the superconducting quantum computers.

Fast Control Gates & Rabi Limits

The Goal: Minimize gate time t_g to perform operations before decoherence occurs.

- **Driving the Transition:** Microwave pulses at the qubit's resonance frequency ($\omega_{mw} = \omega_{01}$) induce **Rabi Oscillations** between $|0\rangle$ and $|1\rangle$ (more on Rabii crosstalk in slide 91)
- **The Rabi Frequency (Ω):** or the speed of rotation is directly proportional to the microwave amplitude A_{mw}:

$$\Omega \propto A_{mw}$$

- **Rotation Angle (θ):** For a specific gate, e.g., CZ entangle gate:

$$\theta = \Omega \times t_g$$

$$\theta = \pi \implies t_g = \frac{\pi}{\Omega}$$

- **The Power Constraint:** Increasing amplitude A_{mw} allows for faster gates, but is strictly limited by the **thermal budget** of the dilution refrigerator (20 mK).

Example Calculation:

- If a single MW cycle is $T_{mw} = 0.2$ ns (5 GHz) and the required gate time is $t_g = 20$ ns:

The rotation requires ≈ 100 microwave cycles to flip $|0\rangle \rightarrow |1\rangle$.

Surface code

Error Correction Code – Surface Code

Surface code:

- Gold standard of error correction code for QEC in industry (Google, IBM) because of its high fault-tolerance threshold ($\sim 1\%$) & physical qubits space connectivity
- *Maps physical complex noises* (e.g. leakage, decoherence) into discrete **bit-flip** (X) and **phase-flip** (Z) errors.
- *Qubits arrangement in 2D space-layout* thus ancilla-qubits extract X and Z syndromes by performing parity checks
- *Faulty gates and readout noises* captured by repeating 2D measurement over time, creating 3D syndrome volume
- *Qubit errors and localized measurement failures* are distinguished with the time dimension (3rd axis)

Surface Code Elements

- **Logical-Qubits:** Encoded information defined by the global topology of the lattice
- **Distance (d):** The minimum number of physical operations required to flip a logical state; determines the error-suppression capability.
- **Stabilizers (Syndrome-Qubits):** The X and Z parity checks performed by ancilla qubits to identify error locations.
- **Boundaries:** The *Rough* and *Smooth* edges of the 2D plane that define the logical X and Z operators.

Logical Qubits

- **Logical qubits:** Software-designed qubits grouped from many physical qubits to act as a single, reliable unit to detect and collect errors
- Important to build **fault-tolerant quantum computers**. Unlike physical qubits that are fragile due to environmental situations, logical qubits use Quantum Error Correction (QEC) to detect and fix errors.

Key components of logical qubits:

- **Data Qubits** Holding the pieces of logical information; total number of data qubits $= d^2$
- **Syndrome or Ancilla Qubits** helper qubits that interact with the data qubits by performing parity checks (stabilizer measurements) to find data qubit errors; total number of ancilla qubits $= d^2 - 1$

Qubit population in a logical qubit depends on the quality of the qubit:

poor quality $\Rightarrow$ more qubits

Flipped qubit correction is performed classically by classical software to detect and keep track of the errors

Adding more qubits to the logical qubit block decreases the logical error rates only if the qubits' error is below the quantum threshold (details in slide 33)

Logical qubits Parameters

- **Logical qubits distance (d)**
  - Defines the logical qubits threshold by the minimum number of physical faults required to manifest as a logical failure;
  - Determining the system's error-suppression capability;
  - The length of the shortest path across the grid of physical qubits that can flip the logical state

- **Logical qubit size (n)** Total number of physical qubits to create a logical qubit. Logical qubit with distance d:
  - $n = 2d^2 - 1$
  - Data-qubits: d^2
  - Ancilla-qubits: $d^2 - 1$

- Logical qubits with distance d can correct up to $\frac{d-1}{2}$ errors and logical qubits with $\frac{d+1}{2}$ many errors change the logical qubits' state.

Surface Code – Stabilizer

- **Stabilizers:** Operators that represent the health condition of the system. Measuring a healthy system $\equiv$ eigenvalue of $+1$: $S|\psi\rangle = |\psi\rangle$
- Error that anti-commutes with stabilizer $\implies$ measurement flips to -1 $\implies$ an error happened!
- **Core Concept:** Surface code logic relies on the duality of syndrome extraction; Z-Stabilizers track the lattice's connectivity for bit-flip (X) errors, while X-Stabilizers track phase-flip (Z) errors.
  - **Bit-Flip Detection (X-errors):** Measured using **Z-Stabilizers**. Since X and Z anti-commute ($XZ = -ZX$), an X-error will flip the eigenvalue of the Z-Stabilizer from $+1$ to -1.

$$Z\underbrace{(X\,|0\rangle)}_{\text{Error}} = Z\,|1\rangle = -\,|1\rangle \implies \text{Syndrome is } -1 \tag{2}$$

  - **Phase-Flip Detection (Z-errors):** Measured using **X-Stabilizers**. In the X-Pauli ($|+\rangle$ & $|-\rangle$) basis, a Z-error acts like a bit-flip, which the X-stabilizer is designed to catch.

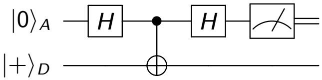
$$X\underbrace{(Z|+\rangle)}_{\text{Error}} = X|-\rangle = -|-\rangle \implies \text{Syndrome is } -1 \tag{3}$$

X-Stabilizer: **Phase-Flip (Z) Error**

To measure Z-Errors on data qubits:

- Reset data and ancilla qubit states: $|\psi(0)\rangle = |0\rangle_A |0\rangle_D$
- Apply Hadamard to data and ancilla change basis (X-basis): $|\psi(0)\rangle = |+\rangle_A |+\rangle_D$
- Apply Z-error on Data only: $|\psi(t_1)\rangle = |+\rangle_A |-\rangle_D$
- Expand basis to $|0\rangle$ and $|1\rangle$:
  $|\psi(t_1)\rangle = \frac{1}{2}(|0\rangle_A |0\rangle_D - |1\rangle_A |1\rangle_D - |0\rangle_A |1\rangle_D + |1\rangle_A |0\rangle_D)$
- Apply CNOT gate: control: ancilla; target: data
  $|\psi(t_2)\rangle = \frac{1}{2}(|0\rangle_A |0\rangle_D - |1\rangle_A |0\rangle_D - |0\rangle_A |1\rangle_D + |1\rangle_A |1\rangle_D)$
- Factoring out common ancilla terms change basis to $|+\rangle$ and $|-\rangle$:
  $|\psi(t_2)\rangle = \frac{1}{2}(|0\rangle - |1\rangle)_A(|0\rangle - |1\rangle)_D = |-\rangle_A |-\rangle_D$
- Apply Hadamard on ancilla: $\psi(t_3) = |1\rangle_A |-\rangle_D$
- Measure ancilla: $|1\rangle_A \Rightarrow |1\rangle_A$

$\Rightarrow$ **The eigenstate is $|1\rangle_A \Rightarrow$ a Phase-Flip error occured!!**

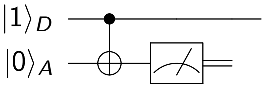


Z-Stabilizer: **Bit-Flip (X) Error**

To measure X-Errors on data qubits:

- Reset data and ancilla qubit states: $|\psi(0)\rangle = |0\rangle_A |0\rangle_D$
- Apply X-error on Data qubit state: $|\psi(t_1)\rangle = |0\rangle_A |1\rangle_D$
- Apply CNOT gate: control: data, target: ancilla $|\psi(t_2)\rangle = |1\rangle_A |1\rangle_D$
- Measure ancilla: $\psi(t_2) = Z_A |1\rangle_A |1\rangle_D = - |1\rangle_A |1\rangle_D$

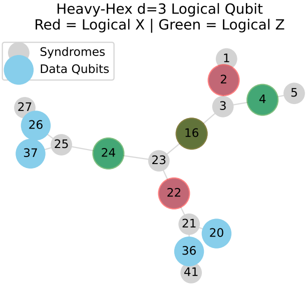
$\Rightarrow$ The eigenstate is -1 $\Rightarrow$ an X-error occured!!



IBM Logical Patch

Extracted logical qubits patch from the IBM-FEZ backend. It uses Heavy-Hex lattice, typically embedded as a zigzag chain of data qubits with ancillas in between. This layout is designed to minimize frequency collisions and crosstalk.

- Logical distance *d*=*3* $\Rightarrow$ 9 data + 8 syndrome qubits, total of 17 qubits
- Extracted logical-X path, the red qubit nodes (from up-to-down): *[2, 16, 22]*
- Extracted logical-Z path, the green qubit nodes (from right-to-left): *[4, 16, 24]*
- Pivot qubit intersect logical-X and logical-Z paths
- Data-qubits: *[2,4,16,20,22,24,26,36,37]*
- Z-Syndromes, bridge-qubits with 2 connections: *[1,5]*
- X-Syndromes, stars-qubits with 3 connections: *[3,21,23,25,27,41]*
- Bit-wise classical allocation for X+Z Syndromes:
  $[0 \rightarrow 1, 1 \rightarrow 5, 2 \rightarrow 3, 3 \rightarrow 21, 4 \rightarrow 23, 5 \rightarrow 25, 6 \rightarrow 27, 7 \rightarrow 41]$



Running Stabilizer Code

Caveat: The logical path defined before is not ideal, because the number of X-Syndromes and Z-Syndromes is not equal

Reminder: Bit-wise classical allocation: [array-index:syndrome-qubit-index] $[0 \rightarrow 1, 1 \rightarrow 5, 2 \rightarrow 3, 3 \rightarrow 21, 4 \rightarrow 23, 5 \rightarrow 25, 6 \rightarrow 27, 7 \rightarrow 41]$

- Running stabilizer for 1024 shots
- No error (noise) injected. The bit-wise Syndrome Outcomes: '00000000': 1024 $\Rightarrow$ **no error injected, no error detected as expected!!!**
- Inject Z-error (Phase-Flip error) on data-qubit 16: Syndrome Outcomes: '00010100': 1024 $\Rightarrow$ error measured in bits 2, 4 corresponding to X-syndromes 3 and 23 connected to data-qubit 16; **expected as X-syndromes 3 and 23 should detect Phase-Flip Error at qubit 16**
- Inject X-error (Bit-Flip error) on data-qubit 4: Syndrome Outcomes: '00000010': 1024 $\Rightarrow$ error measured in bit 1 corresponding to Z-Syndrome 5 connected to data-qubit 4; **expected as Z-Syndrome should detect Bit-Flip Error in data qubit 4**

Errors

Logical Qubits Error

Logical error rate, P_L: Determined by the ratio of the physical error rate, P_{phys}, to the system's fault-tolerance threshold P_{th} (explained in slide 34):

$$P_L \propto \left(\frac{P_{phys}}{P_{th}} \right)^{\frac{d+1}{2}}$$

$P_{phys} < P_{th}$: logical error is suppressed exponentially with d (code distance)
$\implies$ For a system below threshold, increasing the number of physical qubits (increasing d) drastically increases the stability of the logical information.

P_{phys} In a Surface Code cycle, is usually an aggregate of:

- Decoherece (T_1 and T_2 times) (Idle errors)
- Single-Qubit Gate Error: Noise during H or X gates
- Two-Qubit Gate Error: Usually the dominant term (e.g., noise during a CZ gate)
- Measurement/Readout Error
- Qubits' gate measurement in the lab by the Randomized Benchmarking (RB) method (explained in slides 36, 37)

System Fault Tolerant Threshold

Threshold P_{th}:

- Determined by finding the intersection point of logical error curves P_L vs P_{phys} for several logical qubits' distance
- **Definition:** the maximum physical error rate a specific QEC code and decoder can tolerate while providing exponential noise suppression
- Surface Code threshold is $\sim 1\%$ (slide 35)

Computing P_{th} using code-agnostic python-based modules:

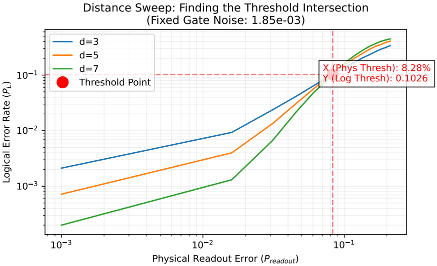
- **Stim:** Stabilizer Simulator that tracks Pauli Frames by simulating the propagation of errors through a circuit. Used to mimic Surface Codes by arranging qubits in a grid and performing specific parity checks
- **Pymatching:** the decoder that takes the detected events and tries to find the errors

Threshold

Parameters

- N_{shots} the number of times `stim` runs the stabilizer and includes Pauli-errors based on the probability given.
  - To allow proper statistical analysis, a considerable number of shots should be given to the simulator
    $P_{phys} = 0.1\%$ & $N_{shots} = 1000 \implies$
    $N_{faulty-event} = P_{phys} \times N_{shots} = 0.001 \times 1000 = 1 \implies N_{shots} > 10,000$
- Distance parameter (d) in `stim` to find P_{th} for various number of qubits
- Range of physical errors that matches the expectation from the datasets

P_{th} is derived from P_L vs P_{phys} plots' intersection for vairous d values



Source: `comp_threshold.py`

Random Benchmark Experiment

Random Benchmark (RB): Experimental protocol to estimate the average error per gate on hardware by measuring how fast a long random sequence of gates loses their fidelity.

RB is robust toward State Preparation And Measurement errors, **SPAM**

IBM backends perform RB sequence:

- Prepare a qubit in the $|0\rangle$ state.
- Select sequence numbers of Clifford gates (such as CNOT, S, H, X), e.g. m = [5,10,20,50]
- Start from l=m[i], and select from Clifford gates as long as *l*, and add the inverse of all the selected gates as l+1 element.
- Select random sequence number, e.g. k =30; each k produces one Clifford gate sequence version
- Each version is repeated for multiple shots, e.g. s = 1240
- Probability of getting the correct state ($|0\rangle$) is averaged over the shots.
- Average the probability over all the *k* versions for the same sequence length to get **P(m[i])**
- Continue this process for each *m[i]* sequence, until *l=len(m)*.

N.B.: As `m` value increases, the probability of a correct state decreases.

RB Fitting

The goal is to derive the average gate Error Per Clifford (EPC) parameter `r`.
Sequence:

- Plot `P(m)` versus `m` decaying curve.
- Fit the curve with:
$$P(m) = A \times \alpha^m + B \tag{4}$$
where A and B are SPAM parameters that only shift the decaying curve.
- Main parameter to measure the error is the **decay parameter:** α
- Criteria for Perfect or Noisy **Hardware**:
  - **Perfect Hardware**: $\alpha = 1$
  - **Noisy Hardware**: $\alpha < 1$
- EPC (Error Per Clifford) parameter is computed from:
$$r = \frac{d-1}{d}(1-\alpha) \tag{5}$$
where d is the dimension parameter computed as $d = 2^n$ and n is the number of qubits per gate. For one-qubit gates, $d = 2$ and $r = \frac{1}{2}(1-\alpha)$

Noises

Hardware Incoherent Noise

Error Discretizations: QEC abstracts analogue hardware noises into discrete digital Pauli errors $\{X, Y, Z\}$ through projective measurement.

Categorizing Incoherent (Random) Hardware Noises:

- **Cosmic Ray (Symptom T_1)** Strike the chip and create phonons' vibrations:
  - Dump qubit's energy
  - Qubits' energy state decays from $|1\rangle$ to $|0\rangle$
  - mapped as a Bit-flip (X) error
- **Magnetic Flux: $1/f$ noise (Symptom T_2)** Causes the qubit's frequency to *jitter* or drift $\Longrightarrow$
  - energy gap fluctuates
  - mapped as a Phase-flip (Z) error
- **Microwave Pulses (Thermal), (Symptom T_1 and T_2)** Thermal energy manifests as random microwave photons bouncing around the cables and the chip.
  - Random photons hit the qubit at random time and phase
  - Causes the qubits to randomly flip (X) or lose its phase (Z)

Key Insight: The stabilizer measurement *snaps* continuous hardware imperfections into digital errors that the decoder can then track and correct

Hardware Control Noise

Control Noise: Coherent/Systematic inaccuracy in the sent signal that manipulates the qubits.

Physical Reasons

- **Pulse imperfection:** Microwave pulses with incorrect amplitude or duration (e.g., rotating 180.5° instead of 180°).
- **Calibration Drift:** Environmental changes (temperature) causing the control electronics to drift *out of tune* slowly
- **Clock Jitter:** Electronics' timing errors that control the pulse start or stop.

Symptoms

- **Gate Errors:** Inaccurate rotations on the Bloch Sphere
- **Readout Errors:** Noise in the signal-processing chain causing state misidentification
- **Leakage:** Frequency *splashes* that push a qubit into a non-computational state ($|2\rangle$)
- **Crosstalk:** Signals *leaking* into neighboring qubits due to frequency crowding

QEC Conversion: Control noise to flipping errors by Projective Measurement to turn analogue noise to digital Pauli flip

- Qubit's state: $|\psi_1\rangle = |0\rangle$
- Control error rotation by θ: $|\psi_2\rangle = cos(\theta)\,|0\rangle + sin(\theta)\,|1\rangle$
- Z-stabilizer + measurement:
  - $|0\rangle$ (No Error) with probability $\cos^2(\theta)$
  - $|1\rangle$ (Bit-Flip / X-Error) with probability $\sin^2(\theta)$

Hardware Control & Incoherent Noise

State Preparation And Measurement (SPAM) occurs at *Preparation* and *Measurement* of a quantum circuit

Hardware errors are usually related to the control electronics and thermal environment rather than the gates.

Physics Reasons

- **State Preparation Errors:** Failure to initialize the qubit into the intended starting state.
  - Thermal Excitation (incoherent)
  - Initialization Pulse Noise (coherent)
  - Mapped as an X-flip (Bit-flip)
- **Measurement Errors:** Failure of the hardware to correctly identify the qubit's final state
  - Signal-to-Noise Ratio (coherent)
  - Decay during Readout (T_1) (incoherent)
  - Crosstalk (incoherent)

Simulation

Heterogeneity Noise

Physical transmon qubits exhibit unique noise profiles that vary significantly across the chip. **Heterogeneous noise** is the unique, real-world error rates of each qubit. It can affect a Surface Code. To study its effect and compare it with a simplified uniform average noise model, online data from IBM backend (Heavy-Hex architecture from *ibm_fez*) is used.
The setup and assumptions of the simulation are as follows:

- **Scalable Geometry:** The code accepts a distance parameter (d) to define the size of the logical surface code patch
- **Syndrome Extraction:** Stabilizer measurements are performed over multiple rounds using CZ-gates and ancilla measurements to detect phase and bit-flip errors using `DEPOLARIZE2` of `stim` python module
- **Error Sources:** The model incorporates specific Readout Errors and Two-Qubit Gate (CZ) errors pulled directly from the backend's daily calibration data
- **Strategic Patch Selection:** The simulation identifies and selects a "Golden Patch"—the subset of physical qubits with the highest fidelity.

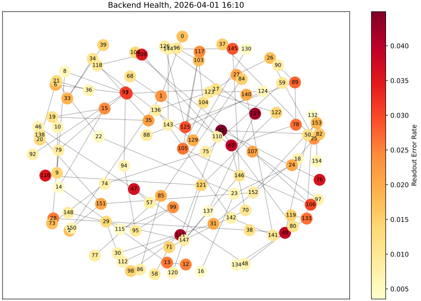
Strategic Patch Selection: The "Golden Patch"

Overview The simulation identifies a high-fidelity subset of qubits on the IBM Fez lattice, resulting in two primary noise scenarios:

- **The "Quiet Zone" (Sub-Average Noise):**
  - **Strategy:** Algorithm identifies and targets high-performance clusters
  - **Action:** Actively filters out "bad apples" (qubits with poor T_1/T_2 or high decoherence)
  - **Result:** Actual Heterogeneous Error $<$ Global Hardware Average
  - **Benefit:** Maximizes logical lifetime by building on the chip's "cleanest" real estate
- **The "Critical Bottleneck" (Super-Average Noise):**
  - **Risk:** High-error outliers occupy critical functional roles (e.g., Data Qubits)
  - **The Math:** In $d = 3$, only 2 failures ($\frac{d+1}{2}$) create a logical bit-flip
  - **Mechanism:** Two data qubits with 4.4% readout error create a "high-probability bridge" for errors
  - **Result:** Actual Logical Error $>$ Uniform Average Model
  - **Impact:** Local "hotspots" dominate the failure rate, pulling the entire patch above the threshold.

Heterogeneity Noise - Simulation

The health map includes single-qubit readout and double-qubit phase-flipping (CZ) errors with distance parameter $d = 3$. This plot was made on April 1, 2026, to make the most recent health map.



By selecting a patch with the fewest errors, the heterogeneous and average Logical Error Rates:

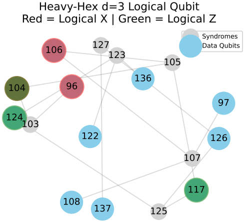
```
Heterogeneous Noise: 0.12556
Average Noise:       0.09273
```

Source: noise_heterogeneity.py

Heterogeneity Noise Map

The four qubits with the highest readout errors, their ID, and their roles are as follows:

```
readout errors (%): 2.612,    2.93,     3.015,   4.224
qubit IDs:          Q105,     Q125,     Q106,    Q127
qubit roles:        Syndrome, Syndrome, Data     Syndrome
```



Speculation:

- **Syndrome Bottlenecks (Q105, Q125, Q127):** Three of the highest-error qubits are syndrome $\Rightarrow$ High readout error can lead to "false flag" detections $\Rightarrow$ the decoder could flip data qubits that were otherwise healthy
- **The Data Risk (Q106):** The error on data qubit, 3.015%, is large. However, as it only affected one data qubit $\Rightarrow < \frac{d+1}{2} = 2$

Source: `plotting.py`

Crosstalk Errors

Definition: The unwanted excitation of a spectator qubit caused by a microwave pulse driving a gate on an adjacent target pair.

Simulation Parameters & Assumptions

- **Worst-Case Coupling:** Selected the CZ gate, as two-qubit operations generate the highest interaction-induced noise
- **Topological Proximity:** Target and spectator qubits are selected as nearest neighbors to simulate noise propagation through the shared coupling bus
- **Control Group:** Qubits are selected from the "Golden Patch" (highest fidelity) to ensure observed errors are strictly crosstalk-driven, not T1/T2 limited
- **Phase Error Mapping:** The crosstalk effect is modeled as a parasitic rotation (θ) on the spectator qubit during the gate time (t_{CZ}): $\theta = 2\pi \times \xi \times t_{CZ}$, where ξ represents the effective coupling strength.
- **Amplified Error Visualization:** The IBM Fez uses tunable couplers with extremely low residual coupling (typically $\xi \approx 1$ kHz). To clearly resolve and demonstrate the crosstalk floor, amplified coupling to $\xi = 100$ kHz to visualize the impact on logical fidelity.

Crosstalk Errors - Simulation Methodology

Objective: To isolate and quantify the crosstalk error induced by a target CZ gate on an adjacent spectator qubit.

Methodology: Randomized Benchmarking (RB)

- **Protocol:** Standard Clifford RB was performed with sequence lengths $m = [1, 10, 50, \ldots, 800]$
- **Error Modeling:**
  - **Isolated Model:** Baseline noise using isolated S_x gate errors
  - **Combined Model:** S_x errors compounded with the parasitic rotation (θ) induced by the target pair's CZ operation
- **Control & Consistency**: Both models utilized identical random seeds and sequence sets to ensure a direct, high-fidelity comparison of survival probabilities

Crosstalk Errors - Data Processing & Fitting

- **Survival Probability (P):** Extracted independently for both isolated and combined sequences
- **Custom Scipy Fitting:** Due to the high-precision nature of the data (small error values), the standard Qiskit fitter was bypassed in favor of a manual scipy.optimize.curve_fit for better convergence
- **Error Per Clifford (EPC):** Calculated using the standard d-dimensional formula ($d = 2$ for single-qubit):

$$EPC = \frac{(d-1)(1-\alpha)}{d}$$

- **Crosstalk Isolation:** The final crosstalk contribution is reported as the delta between the two models:

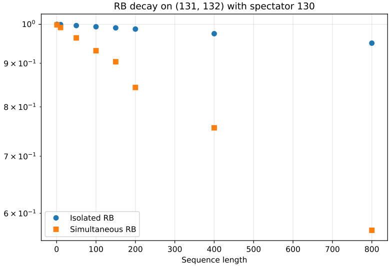
$$EPC_{Crosstalk} = EPC_{combined} - EPC_{isolated}$$

The extracted Error Per Clifford (EPC) for the isolated and simulated crosstalk models reveals a significant shift in logical fidelity:

```
EPC_iso = 5.56e-05, EPC_sim = 4.50e-04
\Delta EPC = 3.94e-04
```

Crosstalk Errors - Results

The isolated and combined RB survival probability versus sequences is shown below:



Key Observation:

- **Divergence Profile:** As expected, the combined survival probability deteriorates rapidly with sequence length (m), confirming that crosstalk dominates the error budget.
- **Differential Validity:** While the absolute survival values include redundant S_x error scaling within the noise model, the subtraction metric ($EPC_{Crosstalk}$) remains a robust estimator for isolating the specific impact of the parasitic CZ rotation.

Source: `xt_leakage.py`

Leakage Errors

Definition: Leakage occurs when a transmon qubit is excited to energy states beyond the computational subspace ($|0\rangle, |1\rangle$) While the anharmonicity, provided by the nonlinear inductance of the Josephson Junction, is designed to isolate the first two levels, high-speed microwave (MW) pulses can still inadvertently drive transitions to higher energy levels, such as the $|2\rangle$ state.
To analyze these effects, this simulation utilizes the following framework:

- **Gate Context:** Analysis is focused on leakage induced during single-qubit gate operations
- **Three-Level System:** The transmon is modeled as a qutrit with states: $|0\rangle, |1\rangle, |2\rangle$
- **Target Operation:** The study simulates a standard X-gate (bit-flip), requiring a precise π pulse calibration.
- **DRAG Compensation:** Used Derivative Removal by Adiabatic Gate (DRAG) technique. This uses a parameter β to scale the derivative of the Gaussian envelope, effectively canceling out-of-subspace (phase) transitions.
- **Parameter Optimization:** Both the MW-pulse duration (T) and the DRAG coefficient (β) are swept to identify the "Leakage Floor" of the system.
- **Pulse Shaping:** The drive utilizes a Gaussian envelope, where the complex component (phase shift) is determined by the β scaled derivative.

Leakage Error - Hamiltonian

The Hamiltonian of the system in the lab frame:

$$H = H_{\text{static}} + H_{\text{drive}}$$

System Properties:

- $H_{\text{static}} = \omega_{01} a^\dagger a + \frac{\alpha}{2} a^\dagger a^\dagger a a$
- $\omega_{01} = 2\pi f_{01}$, and α is the anharmonicity.

Operators (3-level subspace):

$$a = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{bmatrix}, \quad a^\dagger = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{bmatrix}$$

Control Drive (DRAG Pulse): This is the MW pulse of the gate execution

$$H_{\text{drive}} = \text{Re}\left[\Omega(t) e^{i(\omega_d t + \phi)}\right]$$

$$\Omega(t) = I(t) + iQ(t), \quad \text{where } I(t) = Ae^{-\frac{(t-\mu)^2}{2\sigma^2}} \text{ and } Q(t) = \beta \dot{I}(t)$$

Leakage Error - Results

Simulation Setup:

- Solved via `qiskit_dynamics` (`Solver` and `Signal` modules).
- System: $\omega_{01} = 5.0$ GHz, $\alpha = -350$ MHz.
- Pulse: $T = 24$ ns, $\sigma = T/3$, $\mu = T/2$, $A = 0.03$.

Optimization Results:

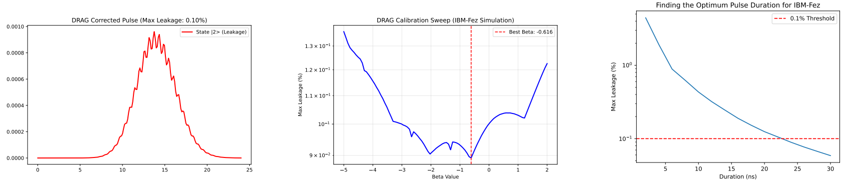


Figure: Left: State populations; Center: β tuning; Right: Leakage vs. Gate duration.

Key Observation

At the selected gate time ($T = 24$ ns), the maximum leakage is **0.1%**, which is on the border of the error threshold. Suggestions to mitigate leakage include increasing the anharmonicity (α) or the pulse width (σ). Without these changes, the only path to lower leakage is active interference via more sophisticated DRAG tuning.

Source: `leakage.py`

Readout Errors

Readout Errors

Objective: As the heterogeneity error computation suggests (slide 46), the readout error is among the highest errors in quantum computing systems. Hence, this study continues with Readout Error Mitigation (REM) techniques as follows:

- **Superconducting readout system**
- **Computational framework**
- **Data visualization**
- **REM techniques**

SC layout and Readout Systems

SC components

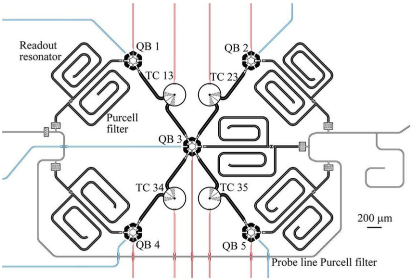
- **Qubits (QB):** Data and Ancilla (Syndrome) qubits
  - same structure
  - Distinct from the software
  - SQUID loops placed between two capacitor planes
- **Resonators:** Cavity connected to the qubits via a capacitor that resonates at a frequency affected by the state of the qubit
- **Purcell Filters:** Acts like a high reflecting mirror that prevents the leakage of the qubits' frequency while passing MW pulses that match the resonator frequency
- **Tunable Coupler (TC):** The SQUID loops, placed between qubits to facilitate double-gate operation while preventing qubits' connection during single-qubit operations (gates, readout, etc)

SC Layout - Continue

Lines: Three classes:

- **FeedLines:** Resonators couple to one single, shared microwave transmission feedline. While the control hardware generates a complex microwave pulse that is a simultaneous combination of all qubits' distinct frequencies
- **DriveLines:** Connected to each qubit separately, changes the state of the qubit as desired (blue lines)
- **FluxLines:** Connected to TC and qubits separately, changes the SQUID energy by shifting the operational frequency (red lines)

Layout example in a 5-qubit QC system:



Resonator Qubit Interaction

Total Hamiltonian of the readout system:

$$H_{tot} = \hbar\omega_r(a^\dagger a + \frac{1}{2}) + \frac{1}{2}\hbar\omega_q\sigma_z + \hbar g(a^\dagger\sigma_- + a\sigma_+)$$

- $\hbar\omega_r(a^\dagger a + \frac{1}{2})$ is the resonator term, and $a^\dagger a$ is the photon number operator
- $\frac{1}{2}\hbar\omega_q\sigma_z$ is the qubit term and σ_z represents the state of the qubit: $|0\rangle, |1\rangle$
- $\hbar g(a^\dagger\sigma_- + a\sigma_+)$ Resonator-qubit interaction:
  - g Vacuum Rabi coupling strength between qubit and resonator from connecting capacitance
  - σ_+ and σ_- Qubit's raising and lowering operators.

Goal: To allow repeatable readout measurement and avoid back action (avoid readout interaction with qubit), **Quantum NonDemolition (QND)** Hamiltonian describes a system where the interaction Hamiltonian commutes with the measured observable (σ_z)
$\Longrightarrow$ Readout measurement should be QND!

$$[H, \sigma_z] = 0$$

QND & Dispersive Hamiltonian

- Condition of dispersive regime: $|\omega_q - \omega_r| = \Delta \gg g$
- Dispersive frequency:
  - Two Level System, QC target: $\chi_{TwoLS} = \frac{g^2}{\Delta}$
  - Three Level System, in leakage (with α as anharmonicity): $\chi_{ThreeLS} = \frac{g^2}{\Delta}\left(\frac{\alpha}{\Delta+\alpha}\right)$
- Dispersive Hamiltonian have three terms: resonator, qubit, and qubit-resonator interaction

$$H_{\text{dispersive}} = \hbar\omega_r\left(a^\dagger a + \frac{1}{2}\right) + \frac{1}{2}\hbar\omega_q\sigma_z + \hbar\chi\left(a^\dagger a + \frac{1}{2}\right)\sigma_z$$

- QND mechanism in action: factor and rearrange dispersive Hamiltonian to see how the resonator and qubit fields entangle with neglecting the only term without operators $(\sigma_z, a^\dagger a)$:

$$H_{\text{dispersive}} = \hbar\left(\omega_r + \chi\sigma_z\right)a^\dagger a + \frac{1}{2}\hbar\left(\omega_q + \chi\right)\sigma_z$$

Closer look at $H_{dispersive}$ terms next slide

Analyzing $H_{dispersive}$

$$H_{\text{dispersive}} = \hbar\left(\omega_r + \chi\sigma_z\right) a^\dagger a + \frac{1}{2}\hbar\left(\omega_q + \chi\right)\sigma_z$$

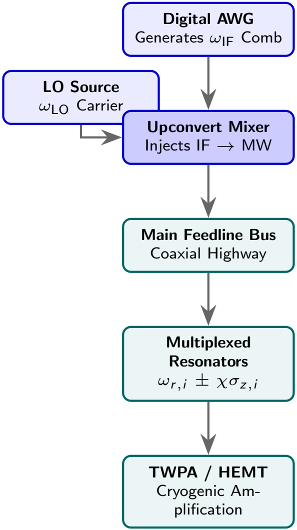
- $\hbar\left(\omega_r + \chi\sigma_z\right) a^\dagger a$ Resonator effective frequency is dynamically pulled by the state of the qubit (σ_z):
  - Qubit's state $|0\rangle$ ($\sigma_z = +1$) $\implies$ resonator frequency: $\omega_r + \chi$
  - Qubit's state $|0\rangle$ ($\sigma_z = -1$) $\implies$ resonator frequency: $\omega_r - \chi$
- Isolating terms with qubit operator $\frac{1}{2}\hbar\sigma_z$, **effective qubit frequency** (ω_q'):

$$\omega_q' = \omega_q + 2\chi a^\dagger a + \chi$$

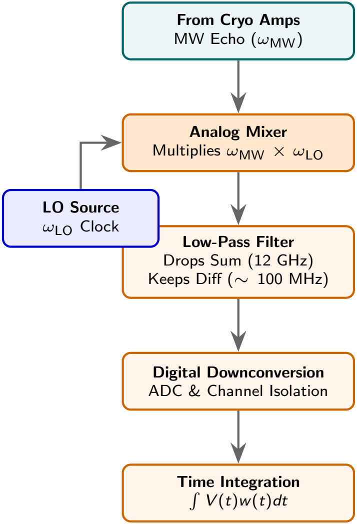
  Photon number operator $a^\dagger a = \hat{n} \implies$ qubit's energy splitting is modulated by two distinct physical effects:
  - **The ac-Stark Shift ($2\chi\hat{n}$):** Every single microwave photon injected to the readout resonator (via the feedline) shifts the qubit's frequency by exactly 2χ. The dynamic shift that depends directly on the measurement drive power
  - **The Lamb Shift (χ):** Empty resonator ($\hat{n} = 0$), shifts qubit's frequency by $+\chi$ $\implies$ Qubit interacts with the vacuum fluctuations (virtual photons) inside the empty resonator cavity.

Hardware: Signal Gen & Cryo

- **Digital IF Generation:** The control hardware synthesizes a low-frequency, multi-tone frequency comb ($\omega_{\text{IF}} \approx 100$ MHz) carrying individual qubit channels.
- **Upconversion Injection:** An analog IQ mixer combines the smart IF pulse with a brute-force carrier wave from a Local Oscillator ($\omega_{\text{LO}} \approx 6$ GHz) to generate the high-frequency MW readout pulse.
- **Frequency Multiplexing:** A single input feedline carries a composite frequency comb $\sum_i A_i \cos(\omega_{r,i} t)$ to address multiple resonators simultaneously.
- **Dispersive Filtering:** The MW pulse travels down the feedline, where targeted resonators shift by $\pm\chi$ based on σ_z, while Purcell filters insulate the qubits from environmental relaxation.



Heterodyne Demodulation Pipeline: DSP



- **Digital Signal Processing (DSP):**

$$x(t) \implies x[t], x[t] = \Sigma_k \delta(t-k) \times x[k]$$

- **Mixing Products:** Multiplying the returning microwave pulse by the LO reference yields:

$$\cos(\omega_{\text{MW}} t)\cos(\omega_{\text{LO}} t) = \frac{1}{2}[\underbrace{\cos(\omega_{\text{MW}} - \omega_{\text{LO}})t}_{\text{Difference (IF)}} + \underbrace{\cos(\omega_{\text{MW}} + \omega_{\text{LO}})t}_{\text{Sum}}]$$

- **Hardware Filtering:** Passing *raw mixed signal* through low-pass filter $\implies$ high-frequency 12 GHz term strips out entirely
- **IF Preservation:** Low-frequency product ($\omega_{\text{IF}} = \omega_{\text{MW}} - \omega_{\text{LO}}$) passes through untouched, locking in the qubit's phase shift at a frequency low enough for clean ADC sampling
- **Integration:** Isolating & integrating digital data paths by channel to output *I* & *Q* voltage values weighted to remove initial and final resonator noises from the integration

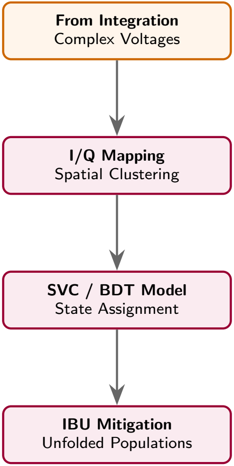
The Analysis Layer: Discrimination & Mitigation

Data Post-Processing Metrology:

- **I/Q Mapping:** Time-integrated voltages resolve into In-phase (I) and Quadrature (Q) components forming distinct state clusters:

$$S = I + jQ$$

- **ML Classification:** Optimized decision boundaries and maximize state discrimination.
- **Error Mitigation:** Mathematical inversion using **Iterative Bayesian Unfolding (IBU)** systematically strips out measurement distortion from the raw confusion matrix.



Raw Readout Voltages

Purpose: Extracting raw readout voltages (I&Q) for qubits' states.

Target system:

- **Processor Architecture:** Selected a spectator pair of physically adjacent qubits on an IBM Quantum Heron processor
- **Crosstalk Modeling:** Pair selection deliberately targets a spectator neighbor with the highest readout error to capture worst-case spectator crosstalk

Experimental Protocol via Qiskit

- **Software Framework:** Executed utilizing the `qiskit_experiments` data acquisition pipeline.
- **State Preparation:** Base states are initialized across the complete two-qubit basis:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

- **Statistical Volume:** Collected 15,000 shots per state to generate a robust, dense dataset of `(I,Q)` voltage coordinate pairs
- **Downstream Utility:** Provides the raw point distributions required to train the SVC/BDT classifiers.

For more details see this Jupyter notebook.

N.B.: The I/Q voltages extracted from `IBM` cloud service are the integrated values due to the account type restriction.

Visualizing Readout Crosstalk on Qubit 1

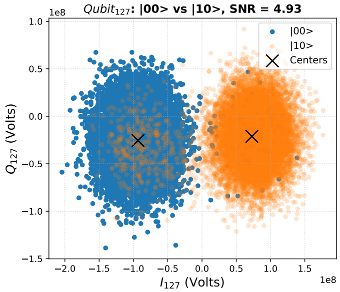
Setup:

- First qubit: `qubit-id=127`
- IBM readout error reported for this qubit $\approx 3\%$
- The Signal-to-Noise-Ratio values printed on the title, computed as:

$$\mathrm{SNR} = \frac{\|\vec{\mu}_1 - \vec{\mu}_0\|}{\frac{1}{2}(\sigma_0 + \sigma_1)}$$

Observable Signatures:

- **Baseline Separation:** Left panel distinct clustering while the neighbor is idle ($|0\rangle$).
- **Centroid Pulling:** Right panel ($|1\rangle$), slight shift in the centers due to coupling.



To see the I&Q distributions individually and their Gaussian fits, see 97

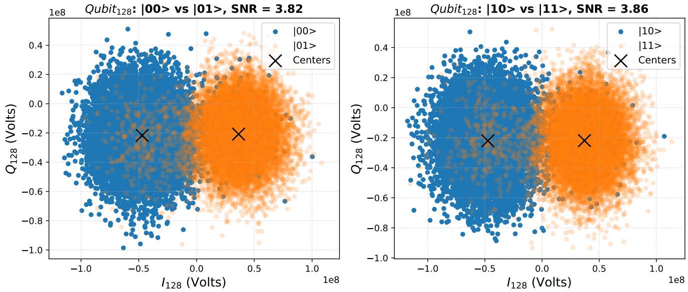
Visualizing Readout Crosstalk on Qubit 2

Setup:

- Second qubit: `qubit-id=128`
- IBM readout error reported for this qubit $\approx 9\%$

Observable Signatures:

- **Baseline Separation:** Left panel clustering while the neighbor is idle ($|0\rangle$), the distinction is less pronounced as the readout error suggests
- **Centroid Pulling:** Right panel ($|1\rangle$), slight shift in the centers due to coupling, the effect of the crosstalk is less obvious $\implies$ the decay time might be the source of higher readout error



To see the I&Q distributions individually and their Gaussian fits, see 99

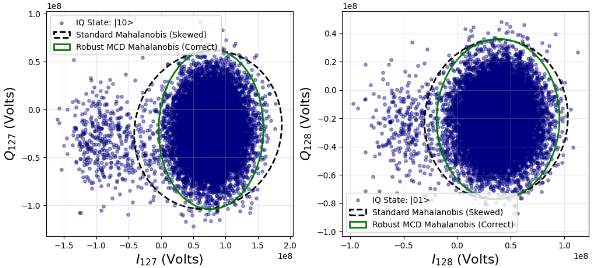
State Discrimination via Mahalanobis Distance

Statistical Distance Metric

- **Covariance Accounting:** Replaces standard Euclidean metrics to account for anisotropic, tilted variance profiles in the I/Q plane (Σ) to remove outliers
- **Metric:** For a measurement vector $\vec{x} = [I, Q]^T$ against a cluster mean $\vec{\mu}$:

$$D_M(\vec{x}) = \sqrt{(\vec{x} - \vec{\mu})^T \Sigma^{-1} (\vec{x} - \vec{\mu})}$$

- **Covariance Estimators:** Evaluated two methods: Robust Minimum Covariance Determinant (MCD) and Empirical Covariance. For details see this
- **Implementation:** MCD and classic Mahalanobis formulations are applied across the full dataset to establish robust cluster boundaries.
- **Result:** The MCD method is more efficient in removing outliers in qubits with lower readout error (`qubit-id=127`)



State Discrimination via Machine Learning

Classification Strategy & Model Architectures

- **Objective:** Maximize multi-qubit state assignment fidelity and map non-linear measurement distortion across the two-qubit manifold.
- **Classification Model:** Evaluated two distinct geometric approaches to classification:
  - *Linear Benchmark:* **Logistic Regression** establishes a baseline and captures mismeasurement symmetric error, such as white noise. (The decision boundary plots in backup slide 104)
  - *Non-Linear Classifiers:* **Boosted Decision Trees (BDT)** and **Support Vector Classifiers (SVC)** capture intricate, distorted cluster structures where manual thresholding fails.
- **Feature Space & Data Volume:**
  - *Input Variables:* Formulated a 4D feature vector per shot: $\vec{X} = [I_1, Q_1, I_2, Q_2]^T$ spanning all 2^2 computational states.
  - *Data Split:* 15,000 shots per state, executed via `scikit-learn` with an 80% training and 20% testing validation split.[1]
- **Decision Boundary Visualization:** Projected 4D boundaries onto individual 2D qubit I/Q planes by systematically holding the spectator qubit's features at $|00\rangle \cup |01\rangle$ subset.

[1] Code implementation details: `github.com/snabili/QEC/test/ml_readout.py`

I&Q Non-Linearity

Purpose: To capture non-linear, state-dependent geometric profiles created by qubit relaxation (T_1) and multi-qubit state-trapping

- **Readout Failure Asymmetry:** The $|1\rangle \rightarrow |0\rangle$ during readout pulse effect on I/Q plane:
  - $\int V(t)dt$ traces a curved trajectory
  - Non-Gaussian distributions (plots show in backup slides 101)
  - Non-linear behavior
- **State-Dependent Coordinate Warp:** The spectator neighbor (1^{st} index in the following items) excited state can shift, stretch, or rotate the primary qubit's clusters unevenly as:
  - $|00\rangle$ **vs** $|01\rangle$**:** ground-state distributions remain isotropic as no energy relaxation can occur from the ground state
  - $|10\rangle$ **vs** $|11\rangle$**:** The excited-state distributions can develop dense, asymmetrical "tails" pointing backward toward the origin, each distorted at a slightly different angle due to the neighbor's cross-resonance coupling[1]

[1] Mathematical representation: $E_{11} - E_{01} \neq E_{10} - E_{00}$

Non-Linear ML Models

- **SVC:**
  - **Radial Basis Function non-linear kernel:** to capture the distribution's outliers
  - **Curved Boundary:** Margin's optimization creates a custom, curved boundary that can bend around the asymmetric relaxation tails
- **BDT:**
  - **Strategy:** Iteratively cutting the 4D feature space into orthogonal hyper-rectangles
  - **Weight Assignment:** In localized I/Q region with a high density of mid-measurement states, the boosting algorithm focuses its subsequent trees
  - **Zone Isolation:** Excited zone, ground zone, and customized step-boundary across the trajectory of the decay tail
- **GMM:**
  - **Gaussian distribution:** Parameters of the $k = 4$ quantum states: Mean (μ_k), Covariance Matrix (Σ_k), Mixing Weight (ω_k)
  - **Probability density function:**

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

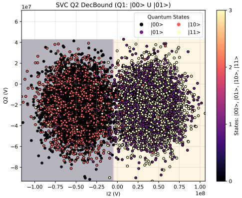
$$\mathcal{N}(x \mid \mu_k, \Sigma_k) = \frac{1}{\sqrt{(2\pi)^d |\Sigma_k|}} \exp\left(-\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1}(x - \mu_k)\right)$$

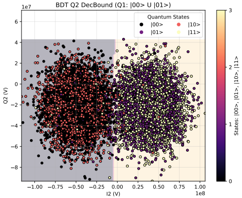
  - **Cluster Shape:** Elliptical; can stretch and rotate to fit elongated data
  - **Assignment Type:** Continuous probabilities

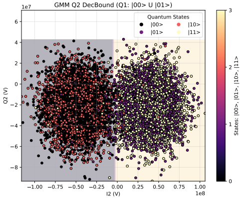
GMM, SVC & BDT Decision Boundaries

Methodology:

- **Dataset:** Train I&Q points for the four (2^2) states: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$
- **Model Optimization** `GridSearch` method to optimize the parameters of the ML models[1]
- **Visualization:** SVC (left) and BDT (right) on test dataset to identify decision boundaries
- **Decision Boundary:** Selected the 2^{nd} qubit conditioned on a definitive state of the spectator qubit to $|00\rangle \cup |01\rangle$ subset[2] to slice 2D plane from 4D hyperplane







[1] Code implementation: `github.com/snabili/QEC/test/ml_gridsearch.py`

[2] More details in backup slides 103:

State Discriminator Performances

Reaout Fidelity as Metric: Computed state discrimination fidelity using:

$$F_{State-Discriminator} = \frac{CorrectlyIdentified}{MisIdentified}$$

The following table shows the readout fidelity of the five discriminators used in this study:

Table: Fidelity metrics across different classification models

State	LR	MAH	GMM	SVC	BDT
$\|00\rangle$	0.962	0.964	0.967	0.965	0.964
$\|01\rangle$	0.958	0.962	0.958	0.961	0.959
$\|10\rangle$	0.956	0.954	0.954	0.954	0.954
$\|11\rangle$	0.944	0.940	0.940	0.940	0.942
State Mean	0.955	0.955	0.955	0.955	0.955

Analyzing the result:

- The readout error (1-Fidelity) reported on the IBM cloud for these two qubits: 127, 128, at the time of this study, was 4% and 9%, respectively, with a mean of 6.5%, resulting in about 2% improvement
- The non-linear ML methods are not outperforming the other linear methods

Linear & Non-Linear Methods

Observation

Linear baselines (LR) match the performance of non-linear classifiers (SVC, GMM, BDT)

Reasons:

- **Geometric Linearity:** The $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ states form distinct clusters in the feature space. Because the natural boundaries are flat, the non-linear freedom of SVC/BDT offers almost no performance advantage.
- **High SNR:** Clean separation between cluster centers, thus the overlapping "fringe" regions are sparse. Changes in boundary curvature only affect a negligible fraction of outlying points.
- **Gaussian Noise Symmetry:** For symmetric Gaussian distributions, the mathematically optimal boundary is inherently linear, aligning with the assumptions of Linear Regression[1]
- **Linear Crosstalk Correction:** A flat hyperplane in the full 4D space ($[I_1, Q_1, I_2, Q_2]$) has enough degrees of freedom to absorb and correct for linear crosstalk between the spectator and target qubits.

Conclusion: Non-linear models will only outperform linear baselines in the presence of system pathologies, such as amplifier saturation, mid-readout relaxation tails, or severe non-linear crosstalk.

[1] Cryogenic amplifier noise is inherently Gaussian

Unfolding in Readout Mitigation

Unfolding

Mathematical techniques to reverse the distorting effects of a noisy measurement apparatus by recovering the true underlying physical distribution from experimental observations

Forward Process ("Folding")

- True quantum states ($\vec{x}$) are scrambled by hardware readout errors and crosstalk.
- The response matrix A mathematically "folds" and mixes distinct features into a noisy, observed ($\vec{y}$) distribution ($\vec{y} = A\vec{x}$).

Inverse Process ("Unfolding")

- Reverses the mixing operation to restore the noisy data to its original components.
- Shared objective of both direct algebra and probabilistic algorithms.

Two Methods, Same Destination

Matrix Inversion (Direct): Undoes the fold in a single linear algebraic step ($\vec{x} = A^{-1}\vec{y}$). Simple, yet sensitive to statistical noise fluctuations

Bayesian Unfolding (IBU): Iterative feedback loop of conditional probabilities to gently unwrap the measurement noise while preserving physical constraints ($0 \leq P \leq 1$).

IBU Method

Objective

Given an observed measurement vector $\vec{y}$ and a known response matrix A ($A_{ji} = P(\text{Measured}_j \mid \text{True}_i)$), find the true state distribution $\vec{x}$

1. **Initialization:** Initial ($n = 0$) uniform prior distribution for $x_i^{(0)}$:

$$x_i^{(0)} = \frac{1}{N_{\text{states}}}$$

2. **Calculate Current Expected Effects:** Project $x_i^{(0)}$ through A_{ji} to get the expected measured distribution:

$$y_j^{(n)} = \sum_i A_{ji} x_i^{(n)}$$

3. **Apply Bayes' Theorem:** Compute $P(T_i|M_j)$:

$$M_{ij}^{(n)} = \frac{A_{ji} x_i^{(n)}}{y_j^{(n)}} = \frac{A_{ji} x_i^{(n)}}{\sum_k A_{jk} x_k^{(n)}}$$

4. **Update True Distribution Guess:** Multiply $M_{ij}^{(n)}$ by actual data counts $\vec{y}_{\text{obs}}$ to yield the upgraded estimate:

$$x_i^{(n+1)} = \sum_j M_{ij}^{(n)} y_{j,\text{obs}}$$

5. **Check Convergence:** Repeat Steps 2–4 for a number of iteration

IBU Implementation

Application

Taking the ML prediction and mitigating the readout error using the IBU method[1].

Methodology:

- **Dataset & ML Model:** Used test data and non-linear ML models SVC (for more results see 105)
- **Response Matrix:** The *scikit-learn* confusion matrix ($CM_{t\times m}$) from ML models for two qubit states
- **CM Elements:** Rows: true/prepared states, columns: measured/predicted counts per states:

$$CM\_\text{scikit-learn}_{t\times m} = \begin{bmatrix} T00M00 & T00M01 & T00M10 & T00M11 \\ T01M00 & T01M01 & T01M10 & T01M11 \\ T10M00 & T10M01 & T10M10 & T10M11 \\ T11M00 & T11M01 & T11M10 & T11M11 \end{bmatrix}$$

- Compute probabilities from confusion matrix elements:

$$P(M_m|T_t) = \frac{CM_{tm}}{\Sigma_t CM_{tm}}$$

Caveat: The difference among scikit-learn CM convention & IBU's method:

$$CM\_IBU_{m\times t} = \begin{bmatrix} T00M00 & T01M00 & T10M00 & T11M00 \\ T00M01 & T01M01 & T10M01 & T11M01 \\ T00M10 & T01M10 & T10M10 & T11M10 \\ T00M11 & T01M11 & T10M11 & T11M11 \end{bmatrix}$$

[1] Code details: `github.com/snabili/QEC/test/ibu_readout.py`

IBU Performance

SVC CM: The IBU performance on the combined fidelity is exceptionally improved. The following table shows the SVC, IBU, and the total counts per state results. The far right-hand side column shows the fidelity before and after IBU application:

State	SVC	IBU	Truth	Fidelity (SVC)	Fidelity (IBU)
$\|00\rangle$	2880	2976	2984	0.965	0.9973
$\|01\rangle$	2812	2916	2925	0.961	0.9969
$\|10\rangle$	2936	3067	3077	0.954	0.9967
$\|11\rangle$	2832	3001	3014	0.940	0.9957

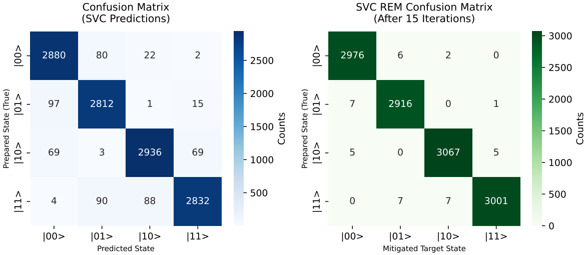


Figure: IBU Performance using SVC Confusion matrix. Left: SVC $CM_{t\times m}$ counts, Right: IBU mitigated counts

Backup Slides

Backup Slides

Smearing Logical States

Smearing Logical States: the *magic* that makes QEC work

- In classical computing, a bit is a single physical spot (e.g., high or low voltage), whereas a single physical spot in quantum computing (e.g., a stray photon) kills the data
- In a Surface Code, CNOT gates are used to entangle multiple Physical Qubits into one Logical Qubit. For instance:
  - Physical State: such as 9 physical qubits ($P_1, P_2, \ldots, P_9$).
  - Logical State: The logical $|0\rangle_L$ not just one qubit (e.g., 0) but a complex, entangled *Superposition of states* involving all 9 qubits.
- Non-Locality: As the information is smeared (**non-local**), a local error (e.g., bit-flip on P_3) doesn't destroy the logical information; instead, it changes the *shape* of the entanglement, which the ancilla qubits can detect and *smooth out*

Entangled gates

Depending on how IBM hardware is set, the entangled gates are presented in the calibration datasets.

Echo-Cross-Resonance setup-CNOT gates:

- Qubits' frequencies are fixed
- Microwave pulse is sent to the control qubit with a frequency equivalent to the target qubit
- Control qubit does not change, but the target qubit flips as the microwave energy tunnels through the control qubit
- if the control qubit is in state $|0\rangle$, the target qubit rotates in one direction, and if it is in $|1\rangle$ state, it rotates in the opposite direction, causing a phase shift.

The SQUID: Enabling Flux-Tunable Couplers

The SQUID: A superconducting loop with two Josephson Junctions sensitive to magnetic flux (Φ).

- **Architecture:** A SQUID-based coupler sits between two *fixed-frequency* transmons. This preserves qubit coherence while allowing on-demand interaction.
- **Mechanism (Frequency Tuning):** Applying magnetic flux to the SQUID shifts the coupler's frequency. This brings the system into a **near-resonance** regime where the $|11\rangle$ state interacts with the non-computational $|20\rangle$ or $|02\rangle$ states.
- **The CZ Gate:** This interaction causes "level repulsion," resulting in a frequency shift Δf. Since Phase $= \int \Delta f \, dt$, we timed the pulse to accumulate a **180° (π) phase shift**.

The "Heron" Advantage

- **Fast Gates:** 30–100 ns operations
- **High Fidelity:** Fixed-frequency qubits avoid flux noise
- **Isolation:** Active couplers eliminate parasitic ZZ crosstalk when "off."

CZ Gate Physics & Energy Level Repulsion

The Mechanism: A flux-controlled interaction between $|11\rangle$ and $|20\rangle$.

- **The State Space:** While we use $|0\rangle$ and $|1\rangle$ for computing, the Transmon has higher levels ($|2\rangle, |3\rangle, \ldots$). A CZ gate utilizes the **non-computational** $|20\rangle$ state as an auxiliary.
- **Avoided Crossing:**
  - As SQUID flux tunes the coupler, the energy of $|11\rangle$ is pushed toward the energy of $|20\rangle$.
  - Quantum mechanics forbids these states from crossing; instead, they **repel** each other (an "avoided crossing").
- **Phase Accumulation:** This repulsion causes a temporary shift in the frequency (Δf) of the $|11\rangle$ state only.

$$\text{Accumulated Phase } \phi = \int_0^{t_g} \Delta f(t)\, dt \tag{6}$$

- **The Result:** By calibrating the pulse duration t_g such that $\phi = \pi$, we achieve the transformation: $|11\rangle \rightarrow e^{i\pi}|11\rangle = -|11\rangle$.

The Leakage Risk

If the pulse is not "adiabatic" (smooth enough), the system may fail to return to $|11\rangle$ and stay in $|20\rangle$. This is a primary source of error in SQUID-based gates.

Bloch Sphere & Density of State

Bloch sphere: The **geometrical representation** of a quantum two-level-system
State configuration and Bloch sphere: Density of a quantum state, $|\psi_i\rangle = \alpha\,|0\rangle + e^{i\varphi}\beta\,|1\rangle$:

- Dirac notation:
$$\rho = \Sigma P_i\,|\psi_i\rangle\,\langle\psi_i| \text{ where } \Sigma P_i = 1 \tag{7}$$
- Matrix notation:
$$\rho = \begin{pmatrix}\alpha^* & \beta^*\end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix} = \begin{pmatrix}\alpha^*\alpha & \alpha^*\beta \\ \beta^*\alpha & \beta^*\beta\end{pmatrix} \tag{8}$$
- Density of state of $|0\rangle\,,|1\rangle$:
$$\rho_{00} = \begin{pmatrix}1 & 0 \\ 0 & 0\end{pmatrix}, \rho_{11} = \begin{pmatrix}0 & 0 \\ 0 & 1\end{pmatrix} \tag{9}$$
    - Diagonal elements: State population
    - Off-Diagonal elements: Coherence
- Coherence indicates how well two states interfere:
    - No coherence, the interference between the two transmons is washing out, or they are not entangled
- **Pure State:** $\rho^2 = \rho$ and $tr(\rho^2) = tr(\rho) = 1$.

Coherent States & Density of States

Caveat Coherence is a property of a single system, entanglement is the property of two qubits states!!!

Example:

- Qubit in ground state at $t = 0$: $\psi(0) = |0\rangle$
- The state evolves at t: $\psi(t) = cos(\theta/2)\,|0\rangle + e^{i\varphi} sin(\theta/2)\,|1\rangle$
- It rotates by $\theta = \pi/2, \varphi = 0, \implies$ it rotates around the x-axis, on the Bloch sphere, the state lies on the equator, the state function: $\psi(t) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- Density of state:

$$\begin{aligned}\rho_{00} &= \frac{1}{2}(|0\rangle + |1\rangle)(\langle 0| + \langle 1|) = \frac{1}{2}(|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| + |1\rangle\langle 1|) \\ &= \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}\end{aligned} \tag{10}$$

State is:

- pure
- coherent
- **Bloch sphere interpretation:** State $|0\rangle$ (south pole) rotated $\theta = \pi/2$ (around x-axis) $\implies$ $|+\rangle$

Coherence Time

- Energy relaxation, T_1 sets the speed limit of the gate.
- T_1 is the energy relaxation, and the time it takes for a qubit before decaying ($|1\rangle \Rightarrow |0\rangle$)
- Qubit's coherence time is theoretically dependent on its material. However, it is practically dependent on the local environment and the microscopic defect, as defects, e.g., residual from fabrication, could have the same frequency as a certain frequency and interact with it, thereby changing the coherence time.
- The error probability due purely to the relaxation time is:

$$Error_{T_1} \approx \frac{T_g}{T_1} \tag{11}$$

where T_g is the gate time.

Phase Coherence

- Phase coherence, T_2, or dephasing time, is how long a qubit can maintain its relative phase in a superposition state
- If T_1 is about energy loss, T_2 is about information loss.
- In the Block sphere, if T_1 is the vertical decay, T_2 is the horizontal smearing around the equator.
- In IBM calibration datasets $T_1 \geq T_2$ for most of the qubits
- The mathematical equation for coherence and dephasing time:

$$\frac{1}{T2} = \frac{1}{2T_1} + \frac{1}{T_\phi} \tag{12}$$

  where T_ϕ phase noise only, T_2 loss of phase coherence in superposition
- Large $T_\phi \Rightarrow T_2 \approx 2T_1$; Small $T_\phi \Rightarrow T_2 \approx T_\phi$ ($T_2 = \frac{2T_1 T_\phi}{2T_1 + T_\phi}$).
- Errors due to dephasing time (T_2): Z Errors;
  Gate Error $= \frac{T_g}{T_2}$, T_g is the gate time.

MW and Coherence time errors

Microwave pulses affect the coherence time errors as changes in their amplitude and phase could cause qubit flipping. MW pulse:

$$V(t) = A(t)cos(\omega_D t + \phi_{Drive}) \tag{13}$$

- **X-error (bit-flip):** The amplitude of the microwave pulse and the area of the pulse envelope affect the coherence time, T_1, . Too long or too hard microwave pulse, thermal relaxation could apply amplitude noise. This will cause θ rotation in the Bloch sphere.
- **Z-error (phase-flip):** The phase shift depends on the phase of the microwave carrier. This will affect the T_2, coherence phase.

T_1 and Readout Length

To estimate how T_1 affects the readout length, the decay probability is computed using:

$$P(decay) = 1 - e^{-t_{meas}/T_1} \tag{14}$$

where t_{meas} is the readout length in (ns). The figure below shows the histogram of the probability of decay divided by the probability of measuring state zero while preparing state one ($P(0|1)$).

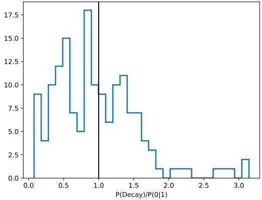


Figure: Decay Probability ratio to $P(0|1)$). for about 43% of data $P(decay) > P(0|1)$

For about 43% of the dataset $P(decay) > P(0|1)$ and this is because the the $P(0|1)$ is averaged over the t_{meas}. If the pulse detects the decay at 1400 ns, then $P(0|1)$ is averaged and given the state as $|1\rangle$

Single Qubit Gate Errors

- S_X is the native one-qubit gate, as it is the native microwave pulse. It rotates the quantum state $\pi/2$ around the X-axis in the Bloch sphere.
- Because S_X is the hardware gate, IBM datasets are calibrated for S_X.
- S_X matrix representation:

$$SX = \frac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} \tag{15}$$

- The other two single-qubit gates are derived from S_x:
  - X-Pauli is $S_X.S_X$, and it is π degree rotated around the X-axis:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{16}$$

  - $R_X(\theta)$ is the rotation around the X-axis with an arbitrary angle (θ):

$$R_X(\theta) = e^{-i\theta/2X} = cos(\theta/2).I - isin(\theta/2).X = \begin{pmatrix} cos(\theta/2) & -isin(\theta/2) \\ -isin(\theta/2) & cos(\theta/2) \end{pmatrix} \tag{17}$$

Rabii Crosstalk

Rabii crosstalk: Unintended Rabii oscillation of a spectator qubit as a target qubit is driven. Important in superconducting quantum systems, defined as the effect of the neighboring qubits as a targeted qubit is excited by some (microwave) pulse to perform a rotation.
The amplitude of the neighboring qubit oscillation is an indication of how strong the crosstalk is.
Reasons:

- Electromagnetic coupling between control lines
- Imperfect in control electronics
- Qubits' closeness

It can cause:

- Unintended rotation
- coherent and dephasing errors
- Gate-dependent noise
- Reduce fidelity of parallel operation

How to mitigate the effect:

- Pulse shaping
- Cancellation tones
- Frequency detuning
- Optimize qubit layout by placing them in heavy-hex layout

Depolarizing Errors

Purpose: The *spherical cow* approximation to simplify the decoder (`PyMatching`) to handle complex physics drifting errors, e.g. coherence time, to discrete X and Z flip errors in a highly symmetric way

Methodology: Takes a perfect quantum state and with a certain probability, p turns it to a state with maximum entropy.

For an initial state ρ, the state after the depolarizing channel, $\epsilon(\rho)$:

$$\epsilon(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \tag{18}$$

For a 2-qubit gate (like a CNOT), a depolarizing error means that after the gate, there is a probability p_2 that a random pair of Pauli operators $\{IX, IY, IZ, XI, XX, \ldots, ZZ\}$ (15 possibilities total) is applied to the two qubits.

By mixing all these possibilities together, the qubit effectively *forgets* its state and points in a completely random direction on the Bloch sphere:

Perfect qubit $\Rightarrow$ Bloch vector length $=1$; Fully depolarized vector $\Rightarrow$ Bloch vector length $= 0$

Coherence Times

Used IBM-fez datasets. Coherence and Dephasing Times

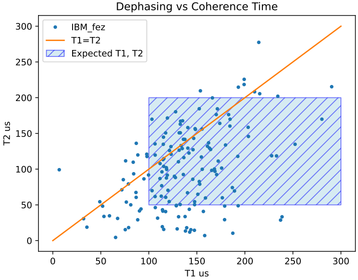


Figure: Dephasing vs Coherence times. Hatch area is the expected value.

Qubits' Measurements

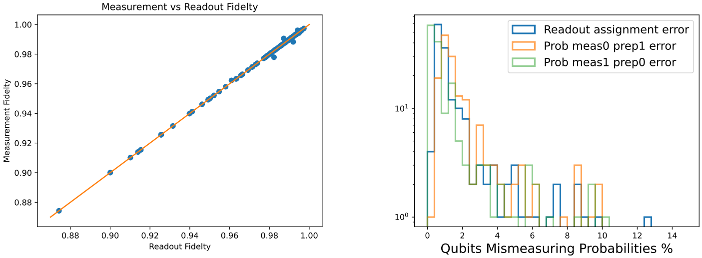


Figure: Measurement vs Readout Fidelities (left) and Mismeasurement probabilities.

Single Qubits' Gate Error

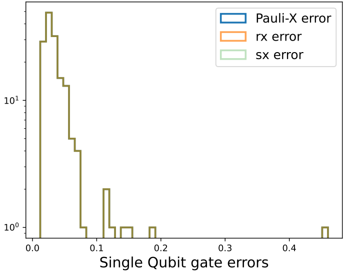


Figure: Single qubit gate errors: $S_x, R_x,$ X-Pauli errors. They have all the same errors.

Double Qubits' Gate Error

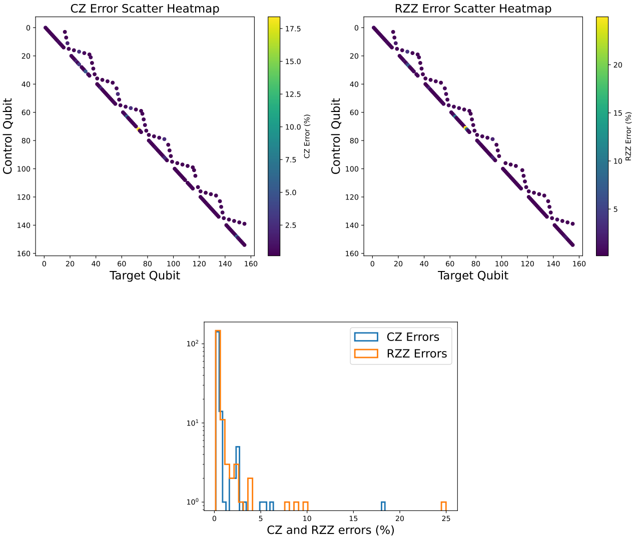
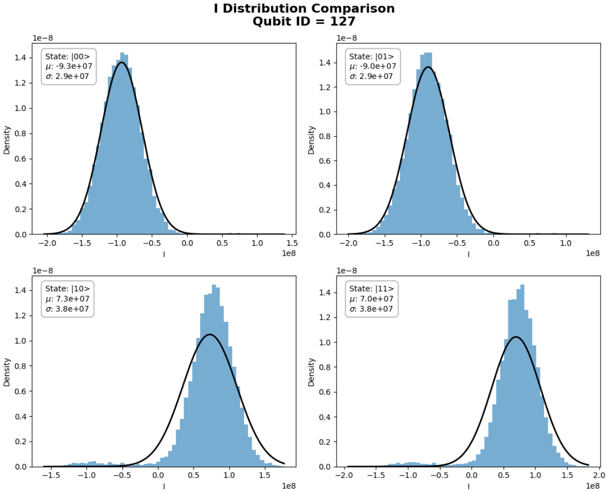
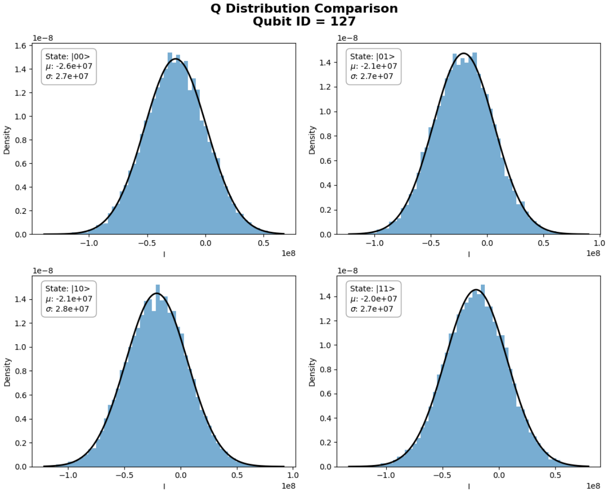


Figure: Single qubit gate errors: C_Z, R_{ZZ} errors. They have all the same errors.

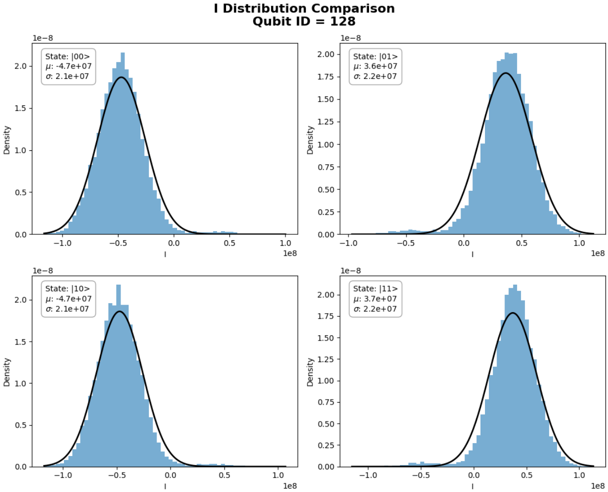
I Distribution- First Qubit



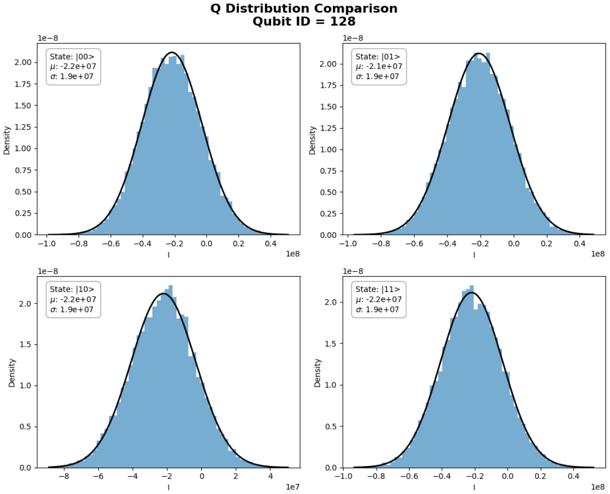
Q Distribution- First Qubit



I Distribution- Second Qubit



Q Distribution- Second Qubit



I Non-Gaussian Distribution

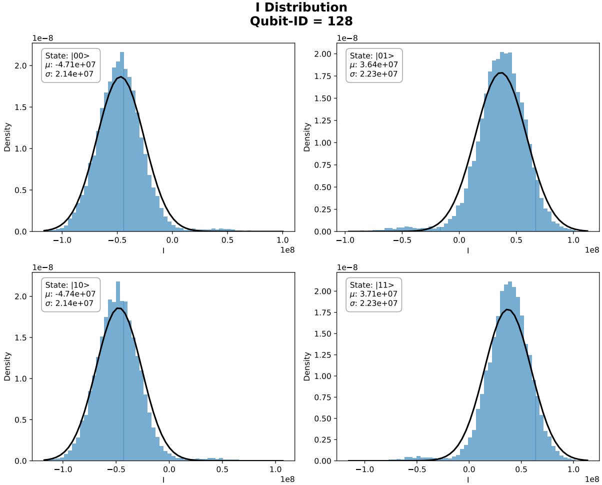


Figure: Four states: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ I distribution with the Gaussian fit. The crosstalk effect is more visual as the neighboring qubit is excited

SVC & BDT Decision Boundaries

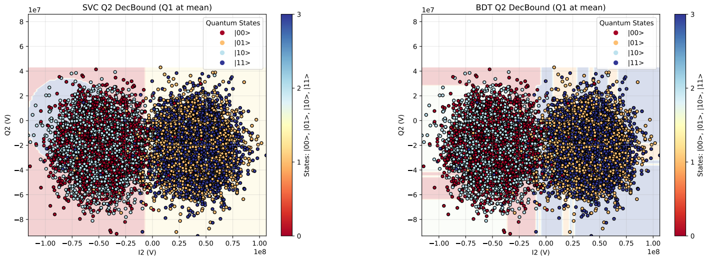


Figure: SVC (left) and BDT (right) of the 2^{nd} qubit's decision boundaries by keeping 1^{st} qubit's $I\&Q$ values at their means. The yellow pocket on the SVC decision boundary indicates a problem.

Origin of the Phantom $|11\rangle$ Decision Boundary

The Problem

When plotting the I_2&Q_2 plane for Qubit 2, an artificial $|11\rangle$ (yellow) decision region appears in the low-density, upper-left quadrant—contradicting the local $|00\rangle$ and $|10\rangle$ data points.

- **High-Dimensional Projection Artifact:** The SVC classifier draws its optimal separating hyperplanes in a 4D feature space: $[I_1, Q_1, I_2, Q_2]$. The plotted background is a flat 2D slice taken across this 4D manifold.
- **The Deception of the Global Mean:** Holding the non-targeted features at their global averages $(\langle I_1\rangle, \langle Q_1\rangle)$ slices the model through an unphysical intermediate space located *between* the true Qubit 1 clusters rather than through a natural state center.
- **Unconstrained Hyperplane Extrapolation:** In low-density data regions (the upper-left quadrant), the SVM's margin-maximization math is unconstrained. Slicing a slightly tilted 4D boundary plane at an unphysical coordinate causes the distant $|11\rangle$ region to mathematically "bleed" into the local I_2–Q_2 perspective.

Mitigation

Condition the slice on a definitive state of the spectator qubit (e.g., conditioning on $\langle I_1, Q_1\rangle$ restricted purely to the $|00\rangle \cup |01\rangle$ subset) to restore physical boundary alignment.

LogisticRegression Decision Boundary

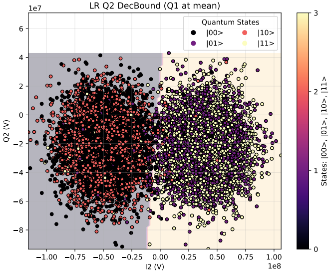


Figure: Logistic Regression decision boundary for the 2^{nd} qubit. The decision boundary is linear as expected.

IBU Performance: GMM & BDT

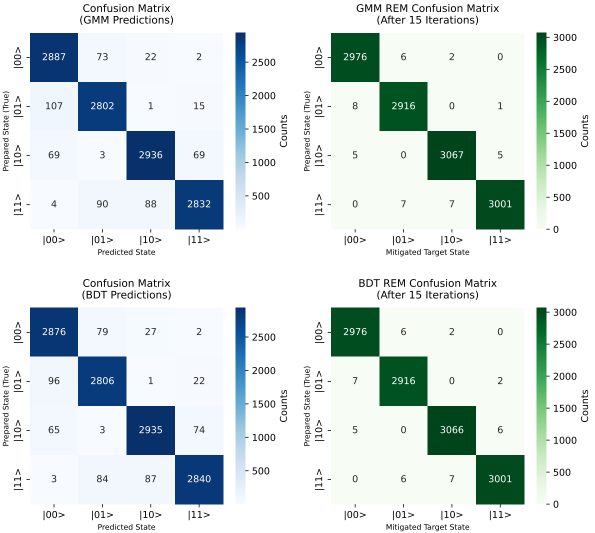


Figure: IBU performance. Left: GMM and Right: BDT confusion matrices

List of Potential Areas of Study

- **Noise model construction:** Used T_1 and T_2, gate errors, and readout errors to study the full simulator of device noise
- **Error-budget analysis:** All error columns to evaluate Dominant error sources
- **CNOT error estimation:** CZ/RZZ + 1Q errors to estimate Effective CNOT fidelity
- **Benchmarking simulation:** Gate errors to study RB, IRB, purity benchmarking
- **Error mitigation:** Readout + gate errors to evaluate the Performance of mitigation methods
- **Decoherence studies:** T_1/T_2 + gate lengths to evaluate Coherence-limited fidelity
- **QEC feasibilities:** All physical error rates to measure Logical error estimates
- **Drift analysis:** Time-series CSVs to evaluate Device stability